Task 6: Plan Analysis

Objective

The objective is to collect every required rock sample and deliver it to Base. The rover must finish at Base, and both storage bays must be empty.

Domain Design

The Mars rover domain uses typed STRIPS actions. The move action follows directed connections between locations. The rover can collect soft rocks directly, while hard rocks require it to pick up the drill at SiteE first. Two storage bays limit the rover to carrying two rocks at a time. The deposit action places a carried rock at the rover's current location and frees its bay. The extended problem closes the direct Base-SiteB route and adds two rocks.

Plan Summary

1. Plan lengths. The shortest plans calculated from the supplied action model contain 25 actions in the base scenario and 40 actions in the extended scenario. Each move, drill pickup, collection, and deposit counts as one action. The supplied files do not include the planner-generated satisficing plans or their step counts, so their exact lengths cannot be verified. A satisficing plan is valid but can be longer than the optimal plan because it may use extra moves or a less efficient collection order. The optimal configuration gives the shortest action count.

2. Storage bays and drill pickup. In the base scenario, the two bays carry Rock2 and Rock4 together on one trip, and Rock5 and Rock3 together on another; Rock1 is delivered separately. The drill is picked up at SiteE before the rover collects the hard rocks Rock5 and Rock3. In the extended scenario, the bays are used for pairs Rock2/Rock4, Rock5/Rock7, and Rock6/Rock3, while Rock1 is delivered separately. The drill is picked up at SiteE during the first trip, before any hard rock is collected. The bays are emptied by depositing their rocks at Base.

3. Extended route after the dust storm. Since the direct Base-SiteB connection is closed, the rover travels from Base through SiteA and SiteC to SiteD, then continues from SiteD to SiteB. The optimal route uses the one-way SiteC-to-SiteD connection. After visiting SiteB, the rover can return to SiteD using the available SiteB-to-SiteD connection, then travel through SiteE, SiteC, and SiteA to reach Base. It does not attempt to travel from SiteD back to SiteC.

Goal-State Checks

The goal is satisfied when all five rocks in the base scenario (or all seven rocks in the extended scenario) are at Base, Bay1 and Bay2 are both empty, and Curiosity is at Base. The goal does not require a separate final condition for the drill; it is picked up during the mission so the rover can collect the hard rocks.